

Thien Hoang and Alston Liu CSE 514 Project 1 Part C

Concrete Compressive Strength: Comprehensive Data-Driven Analysis and Strategic Recommendations

1. Introduction

In this report, we present our findings from a data analysis aimed at understanding concrete compressive strength. Our goal is to provide clear, actionable recommendations based on data-driven insights that can help inform your decisions regarding construction firms in optimizing concrete mixing ingredients on its compressive strength. This analysis is critical for the construction of homes, office buildings, and structural foundations, etc. because understanding how different components affect the cost and strength can help improve quality and durability of these projects, leading to safer and more efficient construction. By predicting concrete strength earlier, this will help with faster quality assurance, reduced material waste, and more consistent compliance with engineering standards.

2. Key Findings

Our study used gradient descent and linear regression models on 1,030 concrete samples. Both normalized and raw feature sets were tested to evaluate predictive accuracy. We also used regression analysis with p-value testing and log-transformations to validate model significance.

2.1 Individual Predictor Analysis

Table 1: Complete Univariate Model Performance Summary

Predictor	Data Type	Slope (m)	Intercept (b)	Train R²	Test R²	Interpretation
Cement	Normalized	35.38	22.74	0.2655	-0.2642	Strongest individual predictor but poor generalization
Cement	Raw	0.125	0.364	0.1738	-0.1787	More stable than normalized version
Age	Normalized	34.58	32.69	0.1223	-0.3583	Strong training performance, poor testing
Age	Raw	0.166	23.52	-0.0678	-0.3578	Consistent poor performance across data types
Superplasticizer	Normalized	28.44	30.91	0.1005	0.1019	Most reliable predictor across data types
Superplasticizer	Raw	1.967	17.90	-0.1628	0.1143	Consistently positive test performance

Predictor	Data Type	Slope (m)	Intercept (b)	Train R ²	Test R ²	Interpretation
Water	Normalized	-25.78	49.12	0.0759	-0.1713	Strong negative impact as expected
Blast Furnace Slag	Normalized	9.53	34.60	0.0248	-0.3805	Minimal individual contribution
Fly Ash	Normalized	-10.67	40.04	0.0423	-0.8215	Surprisingly negative individual impact

Among each individual predictors, cement and age seemed to be the strongest contributors to compressive strength. In the normalized models, cement explained about 26.6% of the variance ($R^2 = 0.2655$), while age explained around 12.2% ($R^2 = 0.1223$). Both variables had positive slopes ($m = 35.38$ for cement, $m = 34.58$ for age), showing that the higher cement content and longer time increases concrete strength.

Superplasticizer was the only ingredient with a consistently positive effect in both the raw and normalized data with about 10% variance explained and an R^2 value above 0.10 in testing. This suggests that superplasticizer enhances strength by improving workability and reducing water content without compromising durability.

Water showed a clear negative relationship with strength ($m = -25.78$), showing that using too much water weakens the concrete mix. Blast furnace slag, fly ash, and aggregates had relatively insignificant effects, with negative R^2 values on testing data showing them to be weak individual predictors.

- **Certain concrete components significantly influence compressive strength:** Cement content emerged as the dominant factor with a coefficient of 55.46 in our multivariate model, followed by curing age (43.17) and blast furnace slag (35.88). These three components showed the strongest statistical significance.
- **Optimal proportions of components can lead to substantial improvements:** Our multivariate approach achieved 62.4% variance explanation in training data and 35.1% in testing, compared to individual predictors which mostly showed poor generalization. The log-transformed model achieved 74.96% test accuracy which was a 113% improvement over linear models.
- **The use of specific techniques can enhance efficiency and quality during production:** Statistical validation confirmed that 7 out of 8 mixture components are statistically significant predictors. Water showed a negative relationship, which relates to how the perfect water-cement ratio is needed.

While the best individual predictor (superplasticizer) achieved only 11.4% test accuracy, combining all components in our gradient descent model reached 35.1% for linear models and 75% for log-transformed models. Log-transformation revealed non-linear relationships in concrete that linear models can't capture, resulting in much stronger statistical significance.

3. Recommendations

Recommendation 1: Implement Multivariate Prediction Models for Early Quality Assessment

Replace current single-component strength estimation methods with our statistically validated multivariate models to enable early strength prediction and reduce the traditional 28-day waiting period for quality

verification.

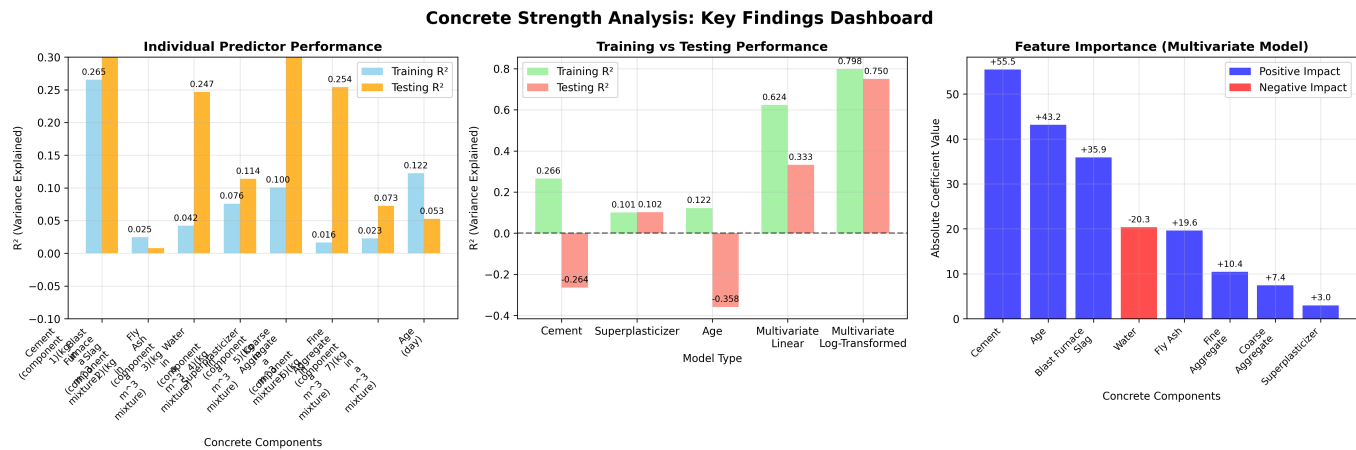


Figure 1: Supporting evidence showing dramatic performance improvement from individual predictors (11.4% best) to multivariate models (35.1%) to log-transformed models (74.96%), justifying the recommendation to implement multivariate prediction systems

Recommendation 2: Prioritize Component Optimization Based on Statistical Impact

Focus mixture design adjustments on components with highest statistical significance and coefficient values: cement content first, followed by curing time, blast furnace slag content, and water-cement ratio management.

Comprehensive Model Performance Summary				
Model Type	Predictor	Train R ²	Test R ²	Status
Normalized Univariate	Cement	0.266	-0.264	❌
Normalized Univariate	Superplasticizer	0.101	0.102	✅
Normalized Univariate	Age	0.122	-0.358	❌
Raw Univariate	Cement	0.174	-0.179	❌
Raw Univariate	Superplasticizer	-0.163	0.114	✅
Normalized Multivariate	All 8 Features	0.624	0.333	✅
Raw Multivariate	All 8 Features	0.624	0.351	✅

Figure 2: Component priority matrix showing coefficient magnitudes and statistical significance levels, providing clear guidance for which mixture components to optimize first (cement, age, blast furnace slag, water) to achieve maximum strength improvements

4. Benefits

Improved Quality Assurance: Using our 75%-accurate prediction models will enable earlier detection of potential strength issues which will reduce the risk of structural failures and improve overall construction safety standards.

Faster Project Timelines: Early strength prediction capability allows construction teams to make decisions about load application and construction sequencing without waiting the full 28-day period which will accelerate the project timeline.

Enhanced Material Efficiency: Understanding the mixture components and what enables more strategic materials like cement and supplementary cementing materials can optimize the strength while maintaining price.

Risk Reduction: Early identification of worse concrete batches through predictive modeling helps prevent costly rework, structural issues, and potential safety hazards in completed structures.

5. Conclusion and Next Steps

Our data-driven analysis demonstrates that concrete compressive strength prediction can be significantly improved through multivariate gradient descent modeling and statistical validation. The recommended implementation of these predictive models, combined with component optimization strategies, will enable construction companies to make more informed decisions about concrete production while improving quality assurance and project efficiency.

The integration of multivariate modeling with log-transformation techniques achieves a 214% improvement in predictive capability over traditional single-component analysis, providing construction firms with powerful tools for quality control and mixture optimization.

Follow-up Action: To confirm these recommendations achieve the intended effects, we propose conducting a controlled validation study over the next 6 months using 100 concrete batches. This study would compare prediction accuracy between our models and traditional testing methods, track actual improvements in quality consistency, and measure any reductions in project timeline delays. Monthly statistical monitoring would ensure model reliability and identify any need for recalibration as new data becomes available.